ELECTRIC CURRENT IN A SIMPLE CIRCUIT

An electric circuit needs a closed path so charge can move. Current is the flow of electrons through that path. A current of 2 amperes means two coulombs of charge pass a point each second.

A cell or battery supplies the energy that pushes the electrons. If the switch is open, the path is broken and the current stops. If the switch is closed, the lamp lights because electrons flow through the filament.

A larger resistance in the same circuit lowers the electron flow. The same 2 ampere current through a thinner wire meets more resistance than through a thicker wire. Keep the unit: 2 A is a current, not a voltage.